HexOct

The Laws of Stacking

Stack equal balls in close-packed layers, each new layer in the hollows of the one below, and the space between them is shared out into a few kinds of piece. This book follows three laws found in this one setting. The Rule of Halves: the gaps of any stacking are built from the gaps of the two pure stackings, whole or in halves. The Swap Rule: when the layers are squashed or stretched, the two mixed pieces always add up to the two pure ones. The Hexagonality Law: every count and every volume of the pieces is a straight line in one number, the share of hexagonal layers.

The laws are tested on real materials, from silicon carbide, ice and battery oxides to spin ice and kagome magnets. The book ends with HexOct: only two tiles, one cubic and one hexagonal, build every close-packed stacking, any tile on any tile; and keyed tiles, HexOct-K, that compute.

The Law gave the pieces, / The pieces prove the law.

DICTO is an artist in Osaka, Japan, with a background in architectural sculpture, stonecarving and Buddhist statuary. He builds geometric sculpture from recycled PET bottles with his own modular systems, B1NR and DICTOspheres.

Open access edition: DICTO DOI 10.5281/zenodo.23731359
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